



UNIVERSITY OF CAPE TOWN
HONOURS PROJECT

Mean-Variance Hedging with Neural Networks using
Heston Model

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*A project submitted in fulfilment of the
requirements for the degree of BBus Sci in
Analytics.*

October 23, 2024



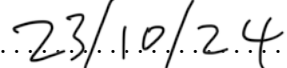
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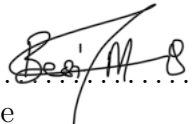
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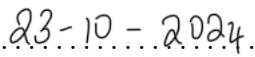
We, Masande Besi and Lwando Mbanguta, declare that this project titled, 'Mean-Variance Hedging with Neural Network using Heston Model' and the work presented in it are our own work. We confirm that:

- This work was done wholly and mainly while in candidature for BBus Sci in Analytics degree at the University of Cape Town.
- Where we have consulted the published work of others, this is always clearly attributed and we have acknowledged all main sources of help.
- Where we have used the words of others, we have indicated this by the use of quotation marks. With the exception of such quotations, this project is entirely our own work.


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Acknowledgements

We would like to extend our heartfelt thanks to our supervisors, Yuri Robbertze and Jake Stangroom, for their invaluable guidance, expertise, and continuous support throughout this project. Their constructive feedback and encouragement have been instrumental in shaping our work. We also express our gratitude to the University of Cape Town for providing the necessary resources and a conducive research environment, making this project possible.

We would like to specially thank Mr Melusi Mavuso for his insightful and engaging Financial Maths classes. His teaching not only deepened our understanding of the subject but also inspired us to pursue more advanced topics in financial modelling. His dedication and passion for the subject have been a significant influence on our academic journey.

Abstract

We investigate the effectiveness of hedging a European basket swap option using a neural network-based hedge compared to a traditional Black-Scholes hedge within a contractual framework involving two parties. The contractual agreement stipulates that at a time T , if the value of the stock owned by party one exceeds that of party two, a trade occurs; otherwise, the option is not exercised. The payoff for the option is given by $\Pi = (S_T^1 - S_T^2)^+$, where S_T^1 and S_T^2 represent the values of the stocks owned by party one and party two, respectively.

To optimise the hedging strategy, we use the Heston model as the underlying path generator, accounting for stochastic volatility, co-variation between stocks and their volatilities, and the correlation between the two stocks. Parameter estimation for the multivariate Heston model is carried out by first discretising the model using Euler's method and then applying Maximum Likelihood Estimation (MLE) on the joint multivariate normal distribution of the stock prices and volatilities. Finally, we implement neural networks to hedge the option and compare the results with a hedge based on the Black-Scholes model.

Keywords: European basket swap option, Neural networks, Black-Scholes model, Heston model, Stochastic volatility, Maximum Likelihood Estimation.

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1 Introduction

In financial risk management, hedging plays an important role in mitigating the uncertainties associated with asset price fluctuations. Traditional hedging techniques, such as the Black-Scholes model, have laid the foundation for modern financial theory, yet they often fall short in capturing the complex dynamics of real-world markets. One of the primary limitations of these classical models is their assumption of constant volatility, which contradicts empirical observations of market behaviour. To address this, the Heston model, introduced by Steven Heston, incorporates stochastic volatility, providing a more realistic representation of asset price movements.

Even with the Heston model, finding the best hedging strategy is still difficult. Mean-variance hedging, which aims to minimise the squared error in hedging, is a common and reliable method. But it traditionally uses linear techniques, which may not fully capture the complex relationships in financial data. Another limitation is the assumption of continuous-time trading in traditional methods, which is often unrealistic due to transaction costs and market liquidity constraints. This is where new developments, like neural networks, come in. Neural networks are good at recognising complex and non-linear patterns, making them a great tool for modelling the complexities of financial markets.

This research aims to explore the integration of neural networks into mean-variance hedging frameworks, using the Heston model as an underlying sample path generator. The Heston model's ability to simulate realistic asset price trajectories provides a robust testing ground for evaluating the performance of neural network-based hedging strategies. The primary objective is to determine whether neural networks can outperform traditional methods in minimising hedging errors and managing financial risk.

To achieve this, we train neural networks on data produced by the Heston model. These networks are built to learn the best hedging strategies that reduce errors over time. The performance of these strategies is then tested and compared with traditional Black-Scholes hedging methods. The study also considers how neural network-based hedging can be used in real financial markets, looking at issues like speed, ease of understanding, and how well the models can adapt to changing market conditions.

In summary, this research aims to show how neural networks can improve mean-variance hedging strategies. By combining advanced machine learning with traditional financial models, the study hopes to offer a new way to manage financial risk in today's complex and volatile markets.

2 Literature Review

The Heston Model, introduced by Steven Heston, has become one of the cornerstones of quantitative finance due to its ability to capture the dynamics of the asset prices and their volatility. It extends the Black-Scholes Model by introducing stochastic volatility, allowing for more accurate modelling of the stock prices, especially in markets with varying levels of volatility.

Numerous studies have applied the Heston Model to price and hedge various types of options, including basket options. For example, [Lin et al. \(2018\)](#) investigated the pricing of basket options under the Heston Model and highlighted the importance of accurately estimating model parameters for effective hedging strategies. Their finding emphasised the need for reliable parameter estimation techniques to improve performance.

Hedging basket options requires sophisticated strategies that account for the co-variation between the underlying stocks and their respective volatilities. The Heston Model offers a flexible framework for designing such strategies. [Xiaoshan \(2020\)](#) examined the effectiveness of the different hedging strategies for basket options using the Heston Model and found that delta hedging, combined with variance swaps, can effectively reduce risk exposure and improve performance.

In the real-world financial markets, contractual agreements often govern the trading and hedging activities of multiple parties. The effectiveness of the hedging strategies can be influenced by the terms of these agreements. [He and Zhu \(2019\)](#) analysed the impact of contractual constraints on the hedging strategies for basket options using the Heston Model. Their study highlighted the importance of aligning hedging strategies with the contractual obligations to maximise hedging effectiveness and minimise associated costs.

Accurate parameter estimation is important for the effective implementation of the Heston Model in option pricing and hedging. Various techniques, such as Maximum Likelihood Estimation (MLE and Monte Carlo simulations, have been employed to calibrate model parameters based on market data. [Chen et al. \(2019\)](#) proposed a novel approach by combining MLE with particle swap optimisation to improve parameter estimation accuracy for the Heston Model, this improved the effectiveness of the hedging strategies for basket options.

While significant progress has been made in understanding the dynamics of option pricing and hedging under the Heston Model, challenges remain in accurately estimating model parameters and designing effective hedging strategies that align with contractual obligations. Further research in these areas is necessary to improve the practical applicability of the Heston Model in real-world financial markets.

3 Theoretical Background

For a comprehensive understanding of derivatives, basket swap options, and the models used in this project, we consulted and reference a range of seminal and recent works. Readers interested in foundational principles of derivative pricing can consult the work of Hull et al. (2009) in *Options, Futures, and Other Derivatives*, which provides an extensive overview of derivative markets and option pricing.

In terms of stochastic volatility models, the Heston model has become a widely cited standard. Heston's original paper, *A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options* Heston (1993), offers a thorough explanation of the model and its implications. For a deeper understanding of extensions and applications of the Heston model, the work by Gatheral (2011) is also recommended.

The Black-Scholes model, which provides the foundation of many option pricing theories, was first introduced in *The Pricing of Options and Corporate Liabilities* by Black and Scholes (1973) and further elaborated in Merton (1973), *Theory of Rational Option Pricing*.

The application of neural networks in finance, particularly in option pricing and hedging, has got increasing interest and recent works as *Deep Hedging* by Buehler et al. (2019) and *Machine Learning for Financial Engineering* by Ruf and Wang (2019) explore the use of neural network for dynamic hedging strategies and provide insights into the advantages of machine learning approaches over traditional methods.

Readers are encouraged to explore these references for a detailed understanding of the models and techniques discussed in the paper. However, the theory of trading in discrete time, the Black-Scholes model, and neural networks as relevant to this project is explained further below.

3.1 Trading in Discrete Time

We begin by developing the theory of financial mathematics in discrete time over a finite horizon. While continuous time models, such as the Black-Scholes framework, are widely used in finance, this project will focus on trading in discrete time. Let $T \in \mathbb{N}^+$ be the horizon and we assume that the trading dates are $t = 0, 1, \dots, T$. We define $\mathbf{S} = (S^1, S^2)$ as a stochastic process representing the prices of two assets. For each $i = \{1, 2\}$, $S^i = \{S_t^i : t = 0, 1, \dots, T\}$ is a discrete-time stochastic process, where S_t^i denotes the price of the asset i at time t .

To account for randomness and the flow of information, we work within a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{F} = \{\mathcal{F}_t : t = 0, 1, \dots, T\}$ represents the filtration capturing the evolution of information over time. We assume that the processes $\mathbf{S}$ are adapted to the filtration $\mathbb{F}$, meaning that their values at each time t are known based on the information available up to that point. Additionally, we assume that Ω is a finite set. The tuple $((\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}), \mathbf{S})$ is referred to as a market.

A trading strategy is a predictable process $\varphi = (\varphi^1, \varphi^2)$, where φ_t^i represents the number of units of asset i to hold from time $t - 1$ to time t . The value of this trading strategy is defined a stochastic process process $V(\varphi) = \{V(\varphi_t^1, \varphi_t^2) : t = 0, 1, \dots, T\}$ defined for $i \in \{1, 2\}$ as:

$$V_t(\varphi) := \sum_{i=1}^2 \varphi_t^i S_t^i \quad t \geq 1 \quad (3.1.1)$$

This represents the portfolio's value as the dot product of the strategy and the asset prices at time t . This is designed to hedge the basket swap option, where the goal is to track the option's price over time by holding a certain number of shares of each stock φ_t^1 and φ_t^2 .

We assume that the hedge portfolio is a self-financing trading strategy such that the only changes to the portfolio's value come from the gains process, there are no external cash flows into or out of the portfolio. The gains process, denoted by $G(\varphi)$, captures the cumulative gains from the trading strategy:

$$G_0(\varphi) := 0, \quad G_t(\varphi) := \sum_{k=1}^t \varphi_k \cdot \Delta S_k \quad \text{for } k = 1, 2, \dots, T. \quad (3.1.2)$$

where $\Delta S_k = S_k - S_{k-1}$. This represents the changes in the portfolio value due to price movements of the assets over time. Thus, the value of the trading strategy

$$V_t(\varphi) = V_0(\varphi) + G_t(\varphi) \quad \text{for } t = 0, 1, 2, \dots, T. \quad (3.1.3)$$

This ensures the portfolio's value evolves solely based on asset price changes and trading decisions, where $V_0(\varphi)$ is the option price at the beginning ($t = 0$). The holding are used to continuously adjust the portfolio, ensuring that the value of the trading strategy $V_t(\varphi)$ tracks the option price at every point in time. The gains process reflects the cumulative adjustments made through the hedging strategy to offset risks from stock price fluctuations. We can represent a trading strategy by specifying an initial value V_0 and predictable holdings in the two assets $\varphi = (\varphi^1, \varphi^2)$ as (V_0, φ) .

A key concept in this framework is arbitrage, where a strategy yields risk-free profits without requiring initial investment. A trading strategy φ is considered an arbitrage strategy if it satisfies the following conditions:

$$V_0(\varphi) = 0, \quad V_T(\varphi) \geq 0 \quad \mathbb{P}\text{-a.s.}, \text{ and } \mathbb{P}(V_T(\varphi) > 0) > 0. \quad (3.1.4)$$

This describes a strategy that starts with zero initial investment, incurs no risk of loss, and yet guarantees a non-negative payoff at maturity with a positive probability of profit. The absence of arbitrage is central to financial theory, as it underpins market efficiency and ensures that prices reflect fair value, removing risk-free profit opportunities.

To model arbitrage-free markets, we introduce the concept of an *equivalent martingale measure* EMM (also called a risk-neutral measure), denoted $\mathbb{P}^*$ on $(\Omega, \mathcal{F})$. This is the probability measure that is equivalent to the real-world measure $\mathbb{P}$, meaning both assign positive probabilities to the same events. A process X is a local martingale under $\mathbb{P}^*$ if X is a local martingale with respect to the filtration $\mathbb{F}$ and the measure $\mathbb{P}^*$. The existence of such a measure is central to the fundamental theorems of asset pricing, ensuring that markets are arbitrage-free and providing a framework for pricing derivatives.

Now, suppose that $\mathbb{P}^*$ is an (EMM) for the price process S , and let φ be a trading strategy such that $V_0(\varphi) = 0$ and $V_T(\varphi) \geq 0$ almost surely. Since $V_t(\varphi)$ is a martingale transform under $\mathbb{P}^*$, and $V_T(\varphi) \geq 0$, it follows that $V(\varphi)$ is also a martingale. By the martingale property, we have $\mathbb{E}_{\mathbb{P}^*}(V_T(\varphi)) = V_0(\varphi) = 0$, which implies that $\mathbb{P}(V_T(\varphi) > 0) = 0$. Therefore, no arbitrage opportunities exist. Conversely, the absence of arbitrage opportunities implies the existence of an EMM. This relationship is known as the *Fundamental Theorem of Asset Pricing I* (FTAP I).

A contingent claim H in a financial market $((\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}), \mathcal{S})$ is defined as a $\mathcal{F}_T$ -measurable random variable, representing a financial payoff that depends on the state of the market at the time T . A trading strategy, φ , is said to replicate the H if the terminal value of the strategy matches the payoff H , almost surely under $\mathbb{P}$:

$$V_T(\varphi) = H \quad \mathbb{P} - \text{a.s.} \quad (3.1.5)$$

In this case, we say that H is attainable or replicable because there exists a strategy that can reproduce the claim. The problem of pricing a contingent claim and the problem of replicating it are deeply connected. If a portfolio φ replicates H , meaning $V_T(\varphi) = H$, and there exists a set of EEM denoted $\mathcal{P}$, then for any $\mathbb{P}^* \in \mathcal{P}$, we have the following relationship:

$$V_0(\varphi) = \mathbb{E}_{\mathbb{P}^*}[V_T(\varphi)] = \mathbb{E}_{\mathbb{P}^*}[H] \quad (3.1.6)$$

This shows that the initial capital required to replicate H , $V_0(\varphi)$, is equal to the expected value of the terminal value H under any risk-neutral measure $\mathbb{P}^*$. Importantly, this initial capital is independent of the choice of the specific EMM $\mathbb{P}^*$. Thus, it makes sense to define the no-arbitrage price of the attainable contingent claim H as the expectation $\mathbb{E}_{\mathbb{P}^*}[H]$ under any EMM $\mathbb{P}^*$, we denote this price as $\pi(H)$.

A financial market $((\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}), \mathcal{S})$, is said to be complete if every contingent claim is attainable. This means that for every contingent claim, there exists a trading strategy that replicates it exactly. A stochastic process X is said to have the *predictable representation property* (PRP) with respect to an *equivalent local martingale measure* (ELMM) $\mathbb{P}^*$, if every $\mathbb{P}^*$ -local martingale M can be expressed as:

$$M_t = M_0 + \sum_{k=1}^t \varphi_k \Delta S_k \quad (3.1.7)$$

The Fundamental Theorem of Asset Pricing II (FTAP II) establishes an important relationship between market completeness, the uniqueness of an ELMM, and the PRP. The theorem states that for a financial market $((\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}), \mathbf{S})$ where *No Free Lunch with Vanishing Risk* (NFLVR) holds, and all components are non-negative, the following are equivalent:

- The market is complete, meaning every contingent claim is attainable.
- There is exactly one ELMM, meaning that there is only one risk-neutral probability measure that prices all assets correctly.
- The process X satisfies the PRP with respect to at least one ELMM

The third property of FTAP II, PRP, provides a replicating strategy for any contingent claim H in a complete market. To explain: Let H be a contingent claim such that $\mathbb{E}_{\mathbb{P}^*}(|H|) \leq \infty$, where $\mathbb{P}^*$ is an ELMM. define the process $M := \{M_t : t = 0, 1, \dots, T\}$, where

$$M_t := \mathbb{E}_{\mathbb{P}^*}(H \mid \mathcal{F}_t) \quad (3.1.8)$$

This is a martingale with $M_T = H$, and by PRP, there exists a predictable process φ^H such that

$$M_t = \mathbb{E}_{\mathbb{P}^*}(H) + \sum_{k=1}^t \varphi_k^H \Delta S_k, \quad t = 1, 2, \dots, T \quad (3.1.9)$$

Substituting $t = T$, we see that the process φ^H is a replicating strategy for H .

In summary, this discrete-time framework allows us to model trading strategies and pricing in an arbitrage-free market through the use of equivalent martingale measures (EMM). The Fundamental Theorems of Asset Pricing (FTAP I and II) provide the necessary conditions for the existence of such measures, ensuring no arbitrage opportunities and market completeness. Specifically, FTAP II demonstrates that in a complete market, every contingent claim can be replicated by a self-financing trading strategy, and there exists a unique EMM that prices all assets correctly.

The Predictable Representation Property (PRP) plays a crucial role in constructing replicating strategies, as it guarantees that any square-integrable martingale can be represented as a stochastic integral with respect to the price process. This allows us to hedge contingent claims by dynamically adjusting our portfolio based on the evolution of asset prices over time.

3.2 Option Pricing with Black-Scholes

In this project, we use the Black-Scholes model to price a basket swap option and explore its use as an alternative hedging strategy, where the hedge ratios are represented as $\varphi = (\varphi_t^1, \varphi_t^2)$. The goal of this section is to explain how the Black-Scholes model is used to price a basket swap

option on two underlying assets, this follows from [Mavuso \(2019\)](#). In a basket swap option, the payoff is based on the difference in performance between the two assets,

$$\Pi = (S_T^1 - S_T^2)^+ \quad (3.2.1)$$

The holder of the option receives a payoff if the value of one asset exceeds the other, based on the final value at maturity.

The Black-Scholes model, introduced by [Black and Scholes \(1973\)](#), provides a mathematical framework for pricing European options in financial markets. It operates under the assumptions of a continuous, complete and arbitrage-free market with constant volatility. The model assumes that the price of a stock follows a stochastic process driven by randomness, which is modelled through geometric Brownian motion.

For a single stock S , its dynamics under this model are captured using a stochastic differential equation (SDE), incorporating two key factors: drift (μ), representing the constant appreciation rate, and volatility (σ), representing the uncertainty or randomness in the stock price. The evolution of the stock price is described by the following SDE:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (3.2.2)$$

Where the initial stock price $S_0 > 0$ and W_t represents a one-dimensional standard Brownian motion on the filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$. To solve this SDE, we apply Itô Lemma and stochastic calculus, the exact solution is given by

$$S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \quad (3.2.3)$$

This solution shows that the stock price follows a log-normal distribution, where the logarithm S_t is normally distributed. To illustrate the martingale property of the stock price under the probability measure $\mathbb{P}$ in the context of arbitrage-free markets we consider a process M_t such that:

$$M_t = e^{\sigma W_t - \frac{1}{2}\sigma^2 t}, \quad \forall t \in [0, T]$$

Then using the properties of Brownian motion and its independent increments we have the following

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}(M_t | \mathcal{F}_u) &= \mathbb{E}_{\mathbb{P}}\left(e^{\sigma(W_t - W_u) - \frac{1}{2}\sigma^2(t-u)} \middle| \mathcal{F}_u\right) + \mathbb{E}_{\mathbb{P}}\left(e^{\sigma W_u - \frac{1}{2}\sigma^2 u} \middle| \mathcal{F}_u\right) \\ &= \mathbb{E}_{\mathbb{P}}\left(e^{\sigma(W_t - W_u)}\right) \left(e^{-\frac{1}{2}\sigma^2(t-u)}\right) + M_u \\ &= \mathbb{E}_{\mathbb{P}}\left(e^{\frac{1}{2}\sigma^2(t-u)}\right) \left(e^{-\frac{1}{2}\sigma^2(t-u)}\right) + M_u \quad \text{since } W_t - W_u \sim \mathcal{N}(0, t-u) \\ &= M_u \end{aligned} \quad (3.2.4)$$

Thus M_t is a martingale under $\mathbb{P}$ with respect to the filtration $\mathbb{F}$. Finally, we decompose the stock price S_t using M_t develops the link for the stock price to possess a martingale property,

$$S_t = e^{\mu t} M_t, \quad \forall u \leq t \leq T$$

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) &= \mathbb{E}_{\mathbb{P}}(e^{\mu t} M_u \mid \mathcal{F}_u) \\ &= e^{\mu t} M_u \\ &= S_u e^{\mu(t-u)} \end{aligned} \tag{3.2.5}$$

From this it becomes clear that the stock price process S is a $\mathbb{P}$ -martingale if and only if $\mu = 0$. In financial markets, this martingale property is key to option pricing under the risk-neutral measure, where the drift μ is replaced by the risk-free rate r .

In financial markets, it is common practice to represent asset prices relative to a single asset—referred to as a *numéraire* asset. This representation allows for easier comparative analysis between assets and simplifies calculations, particularly when evaluating prices in terms of their present value or relative value over time. For the two stocks, labelled $\{1, 2\}$, we can choose one stock as the numéraire, leading to a representation of the discounted prices as follows:

$$S_t^{1/2} = \frac{S_t^1}{S_t^2} \quad (\text{choosing } S^2 \text{ as the numéraire}) \tag{3.2.6}$$

$$S_t^{2/1} = \frac{S_t^2}{S_t^1} \quad (\text{choosing } S^1 \text{ as the numéraire}) \tag{3.2.7}$$

This method of expressing asset prices in terms of a numéraire is particularly useful in the context of option pricing, as it eliminates the need to explicitly account for the risk-free rate in certain computations. Instead, the relative performance of the two assets is directly compared, which simplifies the analysis of their dynamics.

According to Musiela (2022) if the process $S_t^{1/2}$ (the price of stock 1 relative to stock 2) is a martingale under the measure $\mathbb{P}^1$, then the process $S_t^{2/1}$ (the price of stock 2 relative to stock 1) is a martingale under $\mathbb{P}^2$.

We assume that for $u \leq t$, the following holds:

$$\mathbb{E}_{\mathbb{P}^1} \left(\frac{S_t^1}{S_t^2} \middle| \mathcal{F}_u \right) = \frac{S_u^1}{S_u^2} \quad \text{and} \quad \mathbb{E}_{\mathbb{P}^2} \left(\frac{S_t^2}{S_t^1} \middle| \mathcal{F}_u \right) = \frac{S_u^2}{S_u^1} \tag{3.2.8}$$

This assumption states that, under the probability measure $\mathbb{P}^1$, the process $S_t^{1/2}$ (the price of stock 1 relative to stock 2) is a martingale. Similarly, under $\mathbb{P}^2$, the process $S_t^{2/1}$ (the price of stock 2 relative to stock 1) is also a martingale.

To understand this in a practical sense, recall that in an arbitrage-free market, asset prices (or ratios of prices) discounted by the appropriate numéraire should behave as martingales. In this context, the martingale property ensures that, under the respective probability measures $\mathbb{P}^1$ and $\mathbb{P}^2$, the relative prices of the assets evolve in a manner that reflects no arbitrage opportunities.

Since we have established that $S_t^{1/2}$ is a martingale under $\mathbb{P}^1$, it follows by symmetry and the nature of the change of measure that $S_t^{2/1}$ must also be a martingale under $\mathbb{P}^2$. This result is consistent with the fact that when switching the numéraire asset, the relative pricing structure remains martingale-like under the appropriate measure.

To derive the price of the option we introduce a risk-less bank account B in our market such that the market becomes $((\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}), (B, \mathbf{S}))$ and use the bank as the discounting factor i.e. numéraire. The prices for the two assets and the risk-less bank account evolve according to the following SDEs

$$dS_t^i = \mu^i S_t^i dt + \sum_{j=1}^m \sigma_{ij} S_t^i dW_t^j \quad \text{for } i = \{1, 2\} \text{ and } m = \{1, 2, 3\} \quad (3.2.9)$$

$$dB_t = r B_t dt \quad (3.2.10)$$

where μ_i and σ_{ij} are constants and $W = (W^1, \dots, W^m)$ is a m -dimensional Brownian motion process $(\langle W^i, W^j \rangle_t - \delta_{ij}t)$. Equation 3.2.9 can be written succinctly as $dS_t = \mu S_t dt + \sigma dW_t$, for appropriately defined matrices μ and σ . Using the bank asset as numéraire the discounted prices can be expressed as follows

$$X_t^i := \frac{S_t^i}{B_t} \quad \text{and by Itô's lemmas the SDE:}$$

$$dX_t^i = (\mu^i - r) X_t^i dt + \sum_{j=1}^m \sigma_{ij} X_t^i dW_t^j \quad (3.2.11)$$

We assume $\mathbb{F} = \mathbb{F}^W$ and $\mathcal{F} = \mathcal{F}_T$ and attempt to find an ELMM in this model by applying Girsanov's theorem, an outline of this theorem can be found in the Appendix. Let $\gamma = (\gamma^1, \dots, \gamma^m)$ be a W -integrable vector process so that the positive local martingale $\mathcal{E}((\gamma \bullet W))$ is a true martingale. Define the measure $\mathbb{P}^*$ by

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} := \mathcal{E}((\gamma \bullet W))_T = \exp\left(\sum_{i=1}^m \int_0^T \gamma_s^i dW_s^i - \frac{1}{2} \int_0^T \int_0^T \|\gamma_s\|^2 ds\right) \quad (3.2.12)$$

Then $\tilde{W} = (\tilde{W}^1, \dots, \tilde{W}^m)$ is an m -dimensional Brownian motion with respect to $\mathbb{P}^*$, where

$$\tilde{W}_t^i := W_t^i - \int_0^t \gamma_s^i ds, \quad i = 1, \dots, m. \quad (3.2.13)$$

Substituting these equations to the SDE solution for the discounted assets $X_t = e^{-rt}S_t$ we get

$$\begin{aligned} dX_t &= (\mu^i - r)X_t^i dt + \sum_{j=1}^m \sigma_{ij} X_t^i dW_t^j \\ &= \left(\mu^i - r + \sum_{j=1}^m \gamma_t^j \sigma_{ij} \right) S_t^i dt + \sum_{j=1}^m \sigma_{ij} S_t^i d\tilde{W}_t^j \end{aligned} \quad (3.2.14)$$

Thus, for $\mathbb{P}^*$ to be an ELMM, we need to choose γ such that

$$\mu^i - r + \sum_{j=1}^m \gamma_t^j \sigma_{ij} = 0$$

depending on the relationship between m, r, σ_{ij}, μ^i , and the number of stocks involved, this system of equations can have no solution, a unique solution or infinitely many solutions.

Application to two stocks $(S_t^1; S_t^2)$

For two stocks $\mathbf{S} = (S_t^1, S_t^2)$, with their prices have the following dynamics

$$\begin{aligned} dS_t^1 &= \mu_1 S_t^1 dt + \sigma_1 S_t^1 (\rho dW_t^1 + \sqrt{1 - \rho^2} dW_t^2) \\ dS_t^2 &= \mu_2 S_t^2 dt + \sigma_2 S_t^2 dW_t^2 \end{aligned}$$

where W_t^1 and W_t^2 are independent Brownian motion processes and μ^i, σ_i, ρ for $i = \{1, 2\}$ are all constants with $|\rho| < 1$. Define a new process W^3 by

$$W_t^3 := \int_0^t \rho dW_s^2 + \int_0^t \sqrt{1 - \rho^2} dW_s^1 = \rho W_t^2 + \sqrt{1 - \rho^2} dW_s^1 \quad (3.2.15)$$

Then W^3 is also a Brownian motion process with $\langle W^3, W^2 \rangle_t = \rho t$, and the pair of SDEs can be rewritten as

$$dS_t^1 = \mu_1 S_t^1 dt + S_t^1 \sigma_1 dW_t^3 \quad dS_t^2 = \mu_2 S_t^2 dt + \sigma_2 S_t^2 dW_t^2 \quad (3.2.16)$$

To find an ELMM for (S_t^1, S_t^2) , we have the following system of equations

$$\begin{aligned} \mu_2 - r + \gamma_t^1 \sigma_1 \rho + \gamma_t^2 \sigma_2 \sqrt{1 - \rho^2} &= 0 \\ \mu_2 - r + \gamma_t^1 \sigma_2 &= 0 \end{aligned}$$

for γ_t^1 and γ_t^2 , which based on these equations can be seen to have unique solutions.

Under the unique ELMM $\mathbb{P}^*$, the dynamics for the two stocks are given by

$$\begin{aligned} dS_t^1 &= \mu_1 S_t^1 dt + \sigma_1 S_t^1 (\rho d\tilde{W}_t^1 + \sqrt{1-\rho^2} d\tilde{W}_t^2) \\ dS_t^2 &= \mu_2 S_t^2 dt + \sigma_2 S_t^2 d\tilde{W}_t^2 \end{aligned}$$

where $\tilde{W}_t^1$ and $\tilde{W}_t^2$ are $\mathbb{P}^*$ Brownian motions, thus we can again define a third $\mathbb{P}^*$ Brownian motion $\tilde{W}^3$ as

$$\tilde{W}_t^3 := \int_0^t \rho d\tilde{W}_s^2 + \int_0^t \sqrt{1-\rho^2} d\tilde{W}_s^1 = \rho \tilde{W}_t^2 + \sqrt{1-\rho^2} \tilde{W}_t^1 \quad (3.2.18)$$

Under the unique ELMM $\mathbb{P}^*$, the dynamics for the two stocks are given by

$$\begin{aligned} dS_t^1 &= \mu_1 S_t^1 dt + \sigma_1 S_t^1 (\rho d\tilde{W}_t^1 + \sqrt{1-\rho^2} d\tilde{W}_t^2) \\ dS_t^2 &= \mu_2 S_t^2 dt + \sigma_2 S_t^2 d\tilde{W}_t^2 \end{aligned}$$

where $\tilde{W}_t^1$ and $\tilde{W}_t^2$ are $\mathbb{P}^*$ Brownian motions, thus we can again define a third $\mathbb{P}^*$ Brownian motion $\tilde{W}^3$ as

$$\tilde{W}_t^3 := \int_0^t \rho d\tilde{W}_s^2 + \int_0^t \sqrt{1-\rho^2} d\tilde{W}_s^1 = \rho \tilde{W}_t^2 + \sqrt{1-\rho^2} \tilde{W}_t^1 \quad (3.2.19)$$

and rewrite the SDE as

$$dS_t^1 = r S_t^1 dt + S_t^1 \sigma_1 d\tilde{W}_t^3 \quad (3.2.20)$$

$$dS_t^2 = r S_t^2 dt + \sigma_2 S_t^2 d\tilde{W}_t^2 \quad (3.2.21)$$

Where the discount rate r and the volatilities σ_i 's are constant. Solutions to the SDEs are given by

$$S_t^i = S_t^i e^{(r - \frac{1}{2}\sigma_i^2)\tau + \sigma_i(\tilde{W}_T^i - \tilde{W}_t^i)} \quad (3.2.22)$$

$$B_t = B_0 e^{rt} \quad (3.2.23)$$

The logarithms of the stocks S_T^i are normally distributed with the following means and variances:

$$\ln(S_T^i) \sim \mathcal{N}\left(\ln(S_t^i) + \left(r - \frac{1}{2}\sigma_i^2\right)\tau, \sigma_i^2\tau\right) \quad (3.2.24)$$

Under the Black-Scholes model we assert that the arbitrage free price of a contingent claim, H , at time $t \in [0, T]$ is given by

$$\pi_t(H) = S_t^1 \Phi(d_1) - S_t^2 \Phi(d_2) \quad (3.2.25)$$

Where $\tau = T - t$ is the time to maturity for the option and quantities d_1 and d_2 are defined as:

$$d_1 := \frac{\ln\left(\frac{S_0^1}{S_0^2}\right) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad d_2 := d_1 - \sigma\sqrt{\tau} \quad (3.2.26)$$

The combined volatility is defined as $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho_{s^1s^2}\sigma_1\sigma_2}$.

Proof. To prove 3.2.25 we start with the bank account as numéraire under the measure $\mathbb{P}^*$ and apply the change of numéraire technique. The log-normal distribution for $S_T^{i/j}$ for $i, j = \{1, 2\}$, let $X = \ln(S_T^{i/j})$, then $X \sim \mathcal{N}(\mu, \sigma^2\tau)$, where

$$\mu = \ln(S_t^{i/j}) - \frac{1}{2}\sigma^2\tau$$

$$\begin{aligned} \pi_t(H) &= \mathbb{E}_{\mathbb{P}^*} \left[\frac{B_t}{B_T} (S_T^1 - S_T^2)^+ \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{P}^*} \left[\frac{B_t}{B_T} S_T^1 \mathbb{I}_{\{S_T^1 > S_T^2\}} \mid \mathcal{F}_t \right] - \mathbb{E}_{\mathbb{P}^*} \left[\frac{B_t}{B_T} S_T^2 \mathbb{I}_{\{S_T^1 > S_T^2\}} \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_1 \left[\frac{S_t^1}{S_t^1} S_T^1 \mathbb{I}_{\{S_T^1 > S_T^2\}} \mid \mathcal{F}_t \right] - \mathbb{E}_2 \left[\frac{S_t^2}{S_t^2} S_T^2 \mathbb{I}_{\{S_T^1 > S_T^2\}} \mid \mathcal{F}_t \right] \\ &= S_t^1 \mathbb{E}_1 [\mathbb{I}_{\{S_T^1 > S_T^2\}} \mid \mathcal{F}_t] - S_t^2 \mathbb{E}_2 [\mathbb{I}_{\{S_T^1 > S_T^2\}} \mid \mathcal{F}_t] \\ &= S_t^1 \mathbb{P}^1(S_T^1 > S_T^2 \mid \mathcal{F}_t) - S_t^2 \mathbb{P}^2(S_T^1 > S_T^2 \mid \mathcal{F}_t) \\ &= S_t^1 \mathbb{P}^1(1 > S_T^{2/1} \mid \mathcal{F}_t) - S_t^2 \mathbb{P}^2(S_T^{1/2} > 1 \mid \mathcal{F}_t) \end{aligned}$$

The probabilities under the measures $\mathbb{P}^1$ and $\mathbb{P}^2$

$$\mathbb{P}^1(1 > S_T^{2/1} \mid \mathcal{F}_t) = \mathbb{P}(\ln(S_T^{2/1}) < 0 \mid \mathcal{F}_t)$$

This can be written as:

$$\mathbb{P}^1 \left(\frac{S_t^1}{S_t^2} > e^{-\mu + \frac{1}{2}\sigma^2\tau} \right) = \Phi(d_1)$$

where:

$$d_1 = \frac{\ln \left(\frac{S_t^1}{S_t^2} \right) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}$$

Similarly:

$$\mathbb{P}^2(S_T^{1/2} > 1 \mid \mathcal{F}_t) = \Phi(d_2)$$

where:

$$d_2 = d_1 - \sigma\sqrt{\tau}$$

Now, substituting the probabilities $\Phi(d_1)$ and $\Phi(d_2)$ back into the original expression for the price of the contingent claim $\pi_t(H)$, we get:

$$\pi_t(H) = S_t^1 \Phi(d_1) - S_t^2 \Phi(d_2)$$

This completes the derivation of the arbitrage-free price for the option under the Black-Scholes model with two correlated underlying stocks. $\square$

4 Methodology

4.1 Heston Model

The formulas proposed by Steven Heston to describe the movement of the asset prices, where both the asset price and its volatility follow random Brownian motion process.

$$dS_t = \mu S_t dt + \sqrt{\nu_t} S_t dW_t^S \quad (4.1.1)$$

$$dV_t = \kappa(\theta - \nu_t)dt + \xi \sqrt{\nu_t} dW_t^\nu \quad (4.1.2)$$

$$d\langle W^S, W^\nu \rangle_t = \rho dt \quad (4.1.3)$$

where:

- S_t : the asset price at time t .
- ν_t : volatility of the stock price (instantaneous variance).
- μ : risk-free interest rate.
- ξ : variance/volatility of the volatility ν_t (non-negativity).
- θ : long-term price variance.
- κ : rate of reversion to the long-term price variance i.e. the rate at which ν_t reverts to θ (non-negative).
- dt : indefinitely small time increment.
- W_t^S and W_t^ν : Brownian motion for the stock price and its volatility, respectively.
- ρ : the correlation between the Brownian motions, and $\rho \in (-1, 1)$.

By making use of Euler discretisation, we essentially approximate the continuous-time stochastic process of the asset prices and the volatilities into a discrete time framework, this makes it feasible to do analysis using the Heston Model. More importantly, the Heston Model treats the movement of the asset prices as the continuous time process while they occur in discrete time, which is another motivation to discretise the asset price movement.

Moreover, there are two conditions that should be satisfied while calibrating the model.

- The Feller condition: $2\kappa\theta > \xi^2$, this ensures that the instantaneous variance, ν_t , is strictly positive.
- The covariance matrix, Σ_t , is positive definite. This implies that the eigenvalues λ_i for the covariance matrix are strictly positive and $\det(\Sigma_t) > 0$.

4.2 Likelihood Function

The two stocks and the volatilities for $i \in \{1, 2\}$:

$$dS_t^i = \mu^i S_t^i dt + \sqrt{\nu_t^i} S_t^i dt W_t^{S^i} \quad (4.2.1)$$

$$d\nu_t^i = \kappa^i (\theta^i - \nu_t^i) dt + \xi^i \sqrt{\nu_t^i} dt W_t^{\nu^i} \quad (4.2.2)$$

The correlations of the stocks with their volatilities and the correlation between the two stocks:

$$d\langle W^{S^1}, W^{S^2} \rangle_t = \rho_{s^1 s^2} dt \quad (4.2.3)$$

$$d\langle W^{S^i}, W^{\nu^i} \rangle_t = \rho_{s^i \nu^i} dt, \text{ and} \quad (4.2.4)$$

$$d\langle W^{S^i}, W^{\nu^j} \rangle_t = 0 \quad \text{for } i \neq j \quad (4.2.5)$$

Discretise the time interval with Δt such that:

$$S_{t+\Delta t}^i = S_t^i + \mu^i S_t^i \Delta t + \sqrt{\nu_t^i} S_t^i \sqrt{\Delta t} Z_i^S \quad (4.2.6)$$

$$\nu_{t+\Delta t}^i = \nu_t^i + \kappa^i (\theta^i - \nu_t^i) \Delta t + \xi^i \sqrt{\nu_t^i} \sqrt{\Delta t} Z_i^\nu \quad (4.2.7)$$

Now define a multivariate random variable $\mathbf{X}_t = (S_t^1, S_t^2, \nu_t^1, \nu_t^2)$, such that $\mathbf{X}_{t+\Delta t} | \mathbf{X}_t \sim MVN(\mathbf{m}_t, \mathbf{\Sigma}_t)$, where:

$$\mathbf{m}_t = \begin{bmatrix} S_{t-1}^1 + \mu^1 S_{t-1}^1 \Delta t \\ S_{t-1}^2 + \mu^2 S_{t-1}^2 \Delta t \\ \nu_{t-1}^1 + \kappa^1 (\theta^1 - \nu_{t-1}^1) \Delta t \\ \nu_{t-1}^2 + \kappa^2 (\theta^2 - \nu_{t-1}^2) \Delta t \end{bmatrix}$$

and

$$\mathbf{\Sigma}_t = \begin{bmatrix} (\sqrt{\nu_{t-1}^1} S_{t-1}^1)^2 \Delta t & \sqrt{\nu_{t-1}^1} \sqrt{\nu_{t-1}^2} S_{t-1}^1 S_{t-1}^2 \rho_{s^1 s^2} \Delta t & \nu_{t-1}^1 S_{t-1}^1 \xi^1 \rho_{s^1 \nu^1} \Delta t & 0 \\ \sqrt{\nu_{t-1}^1} \sqrt{\nu_{t-1}^2} S_{t-1}^1 S_{t-1}^2 \rho_{s^1 s^2} \Delta t & (\sqrt{\nu_{t-1}^2} S_{t-1}^2)^2 \Delta t & 0 & \nu_{t-1}^2 S_{t-1}^2 \xi^2 \rho_{s^2 \nu^2} \Delta t \\ \nu_{t-1}^1 S_{t-1}^1 \xi^1 \rho_{s^1 \nu^1} \Delta t & 0 & (\xi^1 \sqrt{\nu_{t-1}^1})^2 \Delta t & 0 \\ 0 & \nu_{t-1}^2 S_{t-1}^2 \xi^2 \rho_{s^2 \nu^2} \Delta t & 0 & (\xi^2 \sqrt{\nu_{t-1}^2})^2 \Delta t \end{bmatrix}$$

Thus, the joint probability density function of $\mathbf{X}_{t+\Delta t} | \mathbf{X}_t$ is therefore

$$f(\mathbf{x}_{t+\Delta t} | \mathbf{x}_t) = \frac{1}{(2\pi)^{\frac{4}{2}} |\mathbf{\Sigma}_t|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t)^T \mathbf{\Sigma}_t^{-1} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t) \right] \quad (4.2.8)$$

The likelihood is then defined as the product of the probability density functions of each observation of the random variable. Let $\boldsymbol{\eta} = [\mu^i, k^i, \theta^i, \xi^i, \rho_{s^i s^j}, \rho_{s^i \nu^i}]$, encapsulating all the parameters.

$$\begin{aligned}
L(\boldsymbol{\eta}) &= \prod_{t=1}^n f(\mathbf{x}_{t+\Delta t} | \mathbf{x}_t) \\
&= \prod_{t=1}^n (2\pi)^{-2} |\boldsymbol{\Sigma}_t|^{-\frac{1}{2}} \exp \left[-\frac{1}{2} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t)^T \boldsymbol{\Sigma}_t^{-1} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t) \right]
\end{aligned}$$

The log-likelihood function

$$\log[L(\boldsymbol{\eta})] = -2n \log(2\pi) - \frac{1}{2} \sum_{t=1}^n \log |\boldsymbol{\Sigma}_t| - \frac{1}{2} \sum_{t=1}^n (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t)^T \boldsymbol{\Sigma}_t^{-1} (\mathbf{x}_{t+\Delta t} - \mathbf{m}_t) \quad (4.2.9)$$

4.3 Data Generation

The process of data generation involves randomly generating the stock prices and volatilities of two correlated assets over a one-year horizon, with daily time steps. This generated data serves as the observed stock prices prior to the contractual agreement.

We set the initial parameters for the model as follows:

- Risk-free interest rate (μ): $[0.05, 0.05]$
- Rate of reversion (κ): $[2, 2]$
- Long-term variance (θ): $[0.015, 0.05]$
- Volatility of the volatility (ξ): $[0.1, 0.02]$

Additionally, the correlations are defined as:

- $\rho_{s^1 \nu^1} = 0.3$
- $\rho_{s^2 \nu^2} = 0.3$
- $\rho_{s^1 s^2} = -0.2$

The stock prices and volatilities were initialised at 100 and 0.2, respectively. Figure 1 depicts the simulated stock price trajectories, and the complete code is available in the appendix. With this information of the asset performances, the parties would keenly be interested in understanding the future price paths of the respective stocks, as they impact the payoff structure and the risk profile of the option. Investors can therefore make informed decisions on whether to enter into, hold, or adjust their positions in the basket option, thereby optimising their financial strategies and outcomes.

Prior to entering into the basket swap contract, understanding the potential future stock price paths may prove significant for several reasons. One may aim to hedge against adverse movements in the underlying stock prices and by simulating possible future stock prices, one can

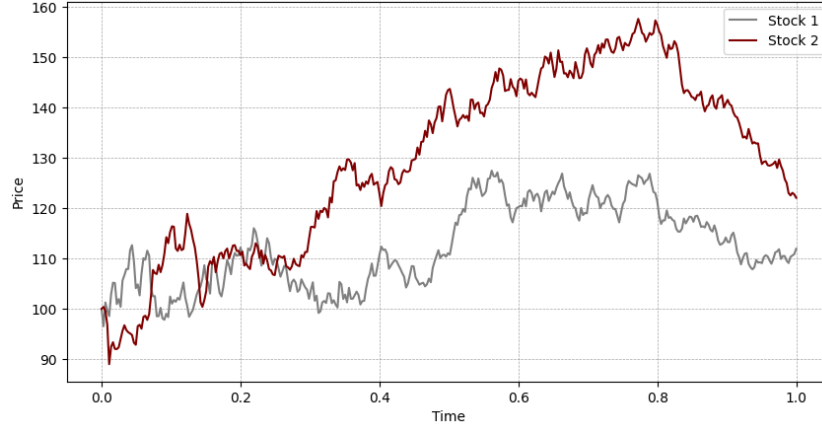


Figure 1: Simulated stock prices under the initial parameters

gain insights into potential scenarios where the stocks may underperform or outperform, allowing them to adjust their risk exposure accordingly. Additionally, a party may have specific

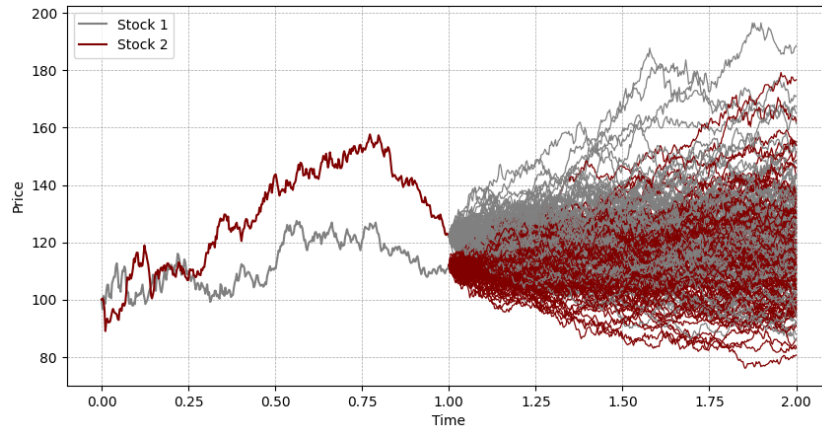


Figure 2: Simulated trajectories of stock prices after $T = 1$, showcasing the range of potential outcomes for basket options.

investment goals or obligations tied to the performance of the underlying assets. By analysing the simulated price paths, they can assess the likelihood of meeting these objectives and make informed decisions regarding the basket swap option.

Given the simulated stock prices in Figure 2, party two can analyse the different scenarios using the trading strategy at which they gain a positive payoff or do not enter into the contract. To visualise these scenarios, the payoff distribution using the simulated future stock prices and the payoff surface are illustrated in Figure 3. The payoff surface shows the different combinations of the respective stock prices whereby party two enters into the swap option and observe a long

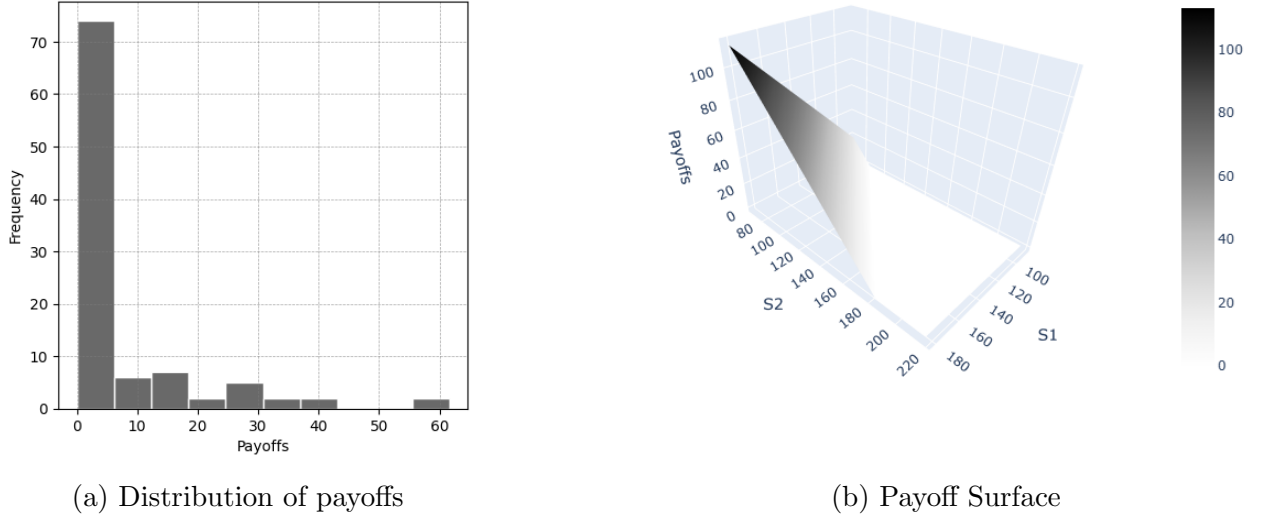


Figure 3: The left sub-figure displays the distribution of payoffs, while the right sub-figure illustrates the payoff surface. These visualisations provide valuable insights for Party two entering into the basket swap contract, aiding in risk assessment.

position, and the scenarios where they do not enter into the swap option. This allows party two to evaluate the potential profitability of entering into the swap option across a wide range of price scenarios. The histogram, shows the distribution of payoffs using the given trading strategy, and the frequency of payoffs if party two enters into the basket swap option. By visualising the frequency and magnitude of payoffs, party two can assess the risk-return profile associated with the option across a range of scenarios.

4.4 Parameter Estimation

In this short section, we explain how to estimate the parameters of the multivariate Heston model from a data set of asset prices. A lot of ground has been covered in the Likelihood Function section. Specifically, we have defined the log likelihood function, which incorporates the key parameters of the model, and we are interested in estimating eleven of them. The parameters of interest;

$$[\mu^1, \mu^2, \kappa^1, \kappa^2, \xi^1, \xi^2, \theta^1, \theta^2, \rho_{s^1 s^2}, \rho_{s^1 \nu^1}, \rho_{s^2 \nu^2}] \quad (4.4.1)$$

These parameters are estimated by maximising the log-likelihood function given the observed data. The estimation process involves initialising the parameters with plausible values and then

using numerical optimisation techniques. The constraints of the parameters ensure that the estimated values adhere to theoretical and practical bounds. The Feller condition ensures the positivity of the variance process, and the determinant of the covariance matrix is maintained within specified bounds to ensure it remains positive semi-definite. This estimation process allows for accurate modelling of asset prices and their volatilities, and enabling the application of the model in various financial analyses and derivative pricing tasks.

5 Hedging: Neural Networks vs Black-Scholes

Neural networks are well-known for their strong ability to predict outcomes and are being used more in finance to find non-linear patterns and give useful insights from data. They are composed of layers of interconnected nodes that process and transform input data to produce predictions. Their flexibility and ability to generalise complex patterns make them effective in dealing with the inherent uncertainties of financial markets.

The Universal Approximation Theorem for neural networks asserts that a feed-forward neural network with a single hidden layer containing a sufficient number of nodes can approximate any continuous function to a desired degree of accuracy, given appropriate activation functions. This powerful property explains their applicability to a wide range of financial problems, where accurate modelling of complex patterns is important.

Neural networks have shown promise in hedging strategies, such as in the work by [Buehler et al. \(2019\)](#), where they outperform traditional methods by offering a more scalable and adaptive approach, especially in capturing complex market dynamics that standard models struggle with. This flexibility and ability to handle large datasets make neural networks a valuable tool in financial decision-making.

5.1 Amazon and Google Historical Data

For this project, we have chosen data from Amazon and Google, and we assume that party one is holding Google stock and is entering into a basket swap option that includes both Google and Amazon stocks. These two stocks have been heavily traded in the past and their market data is readily available, making them ideal for our analysis.

One key reason for choosing these stocks is that the volatility of both stocks is clearly not constant over time. The stock prices have shown significant fluctuations, especially during periods of market instability or company-specific events, as seen in Figure 4. Due to this, the standard Black-Scholes model, which assumes constant volatility, may not provide an accurate representation of the stocks.

The Heston model, on the other hand, is well-suited for this type of situation because it allows for changing volatility over time. This flexibility makes it an appropriate model for handling

varying levels of uncertainty associated with these stocks.



Figure 4: Amazon and Google historical stock prices [Bloomberg LP \(2024\)](#).

The stochastic volatility for both stocks was obtained by first calculating the daily log returns r_t of the stock prices S_t .

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right) \quad (5.1.1)$$

This gives the natural logarithm of the ratio of stock prices from one day to the next. After obtaining the returns, we compute the rolling volatility over a 14-day window using the standard deviation σ_t of the log returns:

$$\sigma_t = \text{std}(r_{t-14:t}) \cdot \sqrt{365} \quad (5.1.2)$$

Here $\text{std}(r_{t-14:t})$ is the standard deviation of returns over the past 14 days, and we multiply by $\sqrt{365}$ to annualise the volatility, converting it into a yearly measure. The rolling volatility gives us a time-varying estimate of how volatile the stock price is over time. The importance of this stochastic volatility lies in capturing the nature of the risk associated with stock prices. Unlike models that assume constant volatility, the Heston model assumes that volatility follows a stochastic process, as seen in equation 4.1.2. This framework allows us to account for time-varying volatility, where the volatility itself is subject to randomness. This feature makes the Heston model better suited for real-world financial data. This historic stochastic volatility for both stocks is displayed in Figure 5.

By leveraging both the observed stock prices and their corresponding volatilities, we apply Maximum Likelihood Estimation (MLE), as described in the Parameter Estimation section, to obtain the multivariate Heston model parameters. MLE is used to estimate the parameters

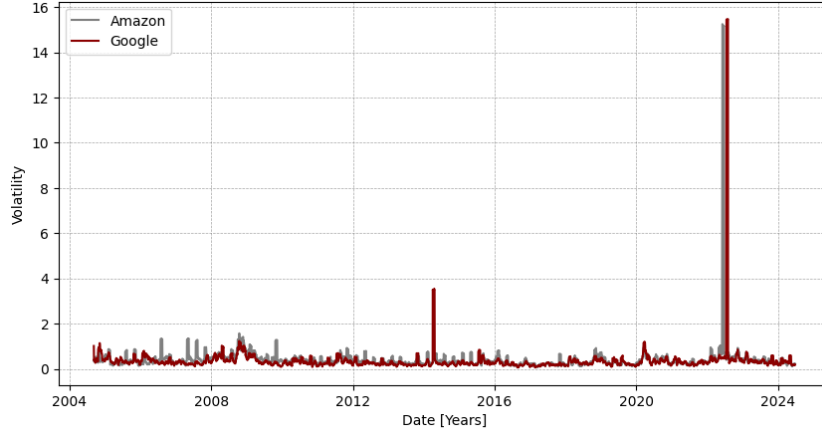


Figure 5: Historic stochastic volatility of Amazon and Google stocks

by fitting the model to the historical price and volatility data, maximising the likelihood of observing the given data under the specified model.

The estimated parameters for the multivariate Heston model are:

$$[\mu^1, \mu^2, \kappa^1, \kappa^2, \xi^1, \xi^2, \theta^1, \theta^2, \rho_{s^1 s^2}, \rho_{s^1 v^1}, \rho_{s^2 v^2}]$$

The estimated values (from the code in Appendix A.2) are:

$$[0.22, 0.268, 4.46, 2.13, 0.179, 0.51, 0.78, 0.42, 0.79, -0.020, -0.022]$$

These values capture the stochastic nature of both stock prices and their volatilities under the Heston framework.

5.2 Stock Price Simulations

For the basket swap option, we assume we are at the end of May, with time to maturity of the option being the end of June. The holder of Google stock, entering into the option, seeks to understand how the two stocks might behave over the month of June. Under the parameters estimated in the previous section using the multivariate Heston model, we simulate many possible future stock price paths across this period.

These simulations are important because they capture a wide range of possible outcomes for stock prices under the stochastic volatility model, accounting for the randomness inherent in both price and volatility. By simulating these paths, the party entering into the contract can observe the potential movements of these stocks and thus, make informed decisions about risk management. Figure 6 illustrates only 100 of these simulated price paths, out of many more that were generated. Each represents as possible scenario for the two stocks over the month of

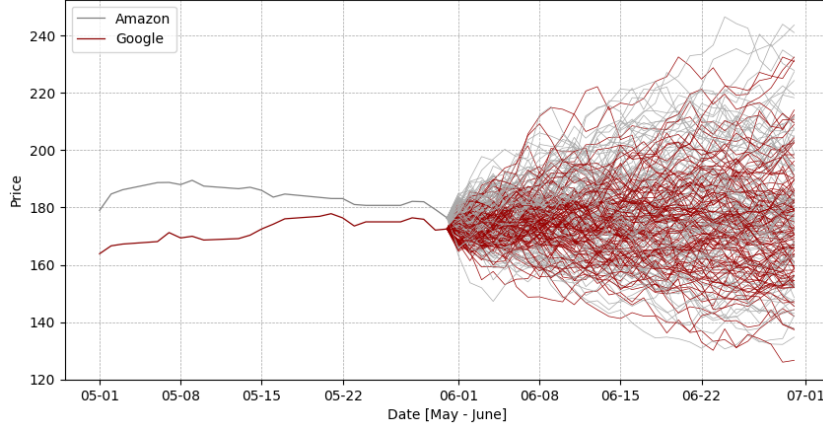


Figure 6: May historic prices and June simulated stock prices under Heston model.

June. The party entering into this contract would like to hedge against all of these scenarios to mitigate the risk associated with unfavourable movements in the stock prices.

5.3 Black-Scholes Hedge

In addition to pricing the option, the Black-Scholes model also provides the hedge ratios for each stock. The hedge ratios, represented as $\boldsymbol{\varphi} = (\varphi^1, \varphi^2)$, define the trading strategy and derived from the partial derivatives of the option price with respect to each stock price [Margrabe \(1978\)](#):

$$\varphi^1 = \Phi(d_1) \quad (5.3.1)$$

$$\varphi^2 = -\Phi(d_2) \quad (5.3.2)$$

These hedge ratios indicate the number of shares of each stock that need to be held in the hedging portfolio to minimise risk. These ratios can be calculated across the month of June to obtain how many shares of Amazon to hold, φ_t^1 , and how many shares of Google to hold, φ_t^2 , at the time t .

The profit and loss (PNL) at the end of the hedging period is the difference between the portfolio value and the option payoff at maturity, ensuring that the trading strategy is designed to closely track the price of the option over its life.

$$\text{PNL} = V_T(\boldsymbol{\varphi}) - H_T \quad (5.3.3)$$

The objective is to obtain a distribution of PNL from the many path simulations, and for mean-variance hedging, we would like for this distribution to be centred around zero with minimal variance. The goal is to obtain a distribution from the many simulated paths that is centred around zero with minimal variance.

5.4 Neural Network Hedge

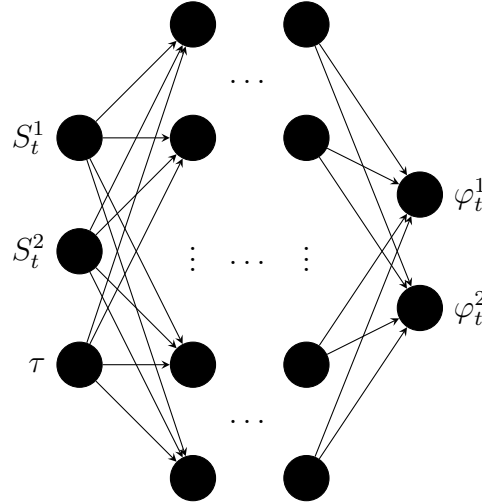
Another way to obtain the hedge ratios is through a neural network. For each path simulation, we use S_t^1, S_t^2 and τ (time to maturity of the option) as inputs to the network to predict the holdings:

$$\varphi^{NN}(S_t^1, S_t^2, \tau) = (\varphi_t^1, \varphi_t^2) \quad (5.4.1)$$

The value of the self-financing portfolio is still obtained using equations (3.1.1) and (3.1.3) and the option price, H_t , is obtained using Black-Scholes formula (3.2.25). The neural network predicts the optimal number of units of each stock to hold at a time t in order to track the option price. The aim is to minimise the difference between the predicted change in portfolio value, based on the trading strategy, and the actual change in the option price.

$$\text{Loss} = \mathbb{E}[(\Delta V_t - \Delta \pi_t)^2] \quad (5.4.2)$$

The goal is to minimise this loss such that the neural network learns to adjust $(\varphi_t^1, \varphi_t^2)$ to effectively hedge the option i.e. the value of the trading strategy at time t , V_t , tracks the option price by matching the change in option price (ΔH_t) at time t .



Input Layer Hidden Layer(s) Output Layer

Figure 7: A generalized standard feed-forward neural network structure for this problem.

Figure 7 illustrates a generalised structure for a standard feed-forward neural network excluding the bias terms. For neural network to predict the optimal holdings for each stock and minimise the loss function, we perform hyperparameter tuning for following;

- Number of hidden layers for the neural network.

- Number of nodes in each hidden layer.
- Activation functions to be used on each hidden layer.
- Optimiser for backpropagation, focusing on Stochastic Gradient Descent (SGD) and Adaptive Moment Estimation (Adam).
- Regularisation methods, choosing between ridge or lasso.

While there are many other hyperparameters that can be tuned, such as the learning rate for backpropagation, we use a scheduler in PyTorch that adjusts the learning process. The output layer makes use of a hyperbolic tangent function to allow for short-selling such that $\varphi_t^i \in [-1, 1]$ for each stock $i \in \{1, 2\}$.

As [Robbertze \(2021\)](#) put it, in machine learning it is standard procedure to split the dataset into a training and validation sets which one trains a model on the training data to evaluate in-sample performance, while the validation dataset is used to assess the model's out-of-sample performance. This approach helps obtain a model that performs well on unseen data and can be used for predictions. Due to lack of data constraints that we are faced with in the financial industry we simulate data for training and likewise simulate data to be used for validation.

Ultimately, our objective is to obtain a distribution of PNL (5.3.3) and compare this distribution to that obtained with Black-Scholes model in the previous section. We aim for a PNL distribution centred around zero with minimal variance, thereby achieving mean-variance hedging. The Black-Scholes hedge relies on a well-established formula that is deterministic and based on the assumption of constant volatility and normally distributed returns. On the other hand, the neural network approach is more flexible, allowing for the learning of complex, non-linear relationships in the data. This can potentially lead to better hedging performance under more realistic market conditions.

6 Results and Analysis

This section, presents the results of our analysis comparing the hedging performance of the Black-Scholes model and the neural network approach for a basket swap option involving Amazon and Google stocks. We aim to assess the profit and loss (PNL) distributions generated by each approach, focusing on their effectiveness in minimising risk.

6.1 Neural Network Training

To train a neural network to hedge, data preparation involved simulating training, validation dataset. Each path simulation contains a time series data for the two assets S_t^1 and S_t^2 , the time to maturity of the option τ , and the corresponding option prices H_t .

The architecture of the neural network consisted of three fully connected hidden layers. Each hidden layer with an activation function based on optimal selections from hyperparameter tuning, while the output layer used a hyperbolic tangent function to allow for the possibility of short-selling. To ensure optimal model performance, we conducted hyperparameter tuning using the Optuna framework. The objective was to get hyperparameters which were used to build and train a neural network model and this was done by selecting the best combination of the following parameters: hidden layer sizes, activation functions for each layer, and the type of regularisation (L_1 or L_2).

After conducting 10 trials with Optuna, the best-performing hyperparameters were identified based on the lowest loss value, Table 1 displays the grid search.

Trial	Loss	Hidden Sizes	Activation Functions	Lambda Reg	Reg
0	4.51	[345, 334, 352, 174, 152, 291]	[Tanh, Sigmoid, Sigmoid, Sigmoid, ReLU, Sigmoid]	0.8789	L_1
1	1.18	[486, 479, 404]	[Sigmoid, ReLU, Sigmoid]	0.0721	L_2
2	1.39	[181, 418, 379]	[Tanh, ReLU, Tanh]	0.8447	L_2
3	1.47	[148, 327, 180]	[Tanh, Tanh, Tanh]	0.6693	L_2
4	2.84	[110]	[Sigmoid]	0.8896	L_1
5	45.24	[438, 351]	[ReLU, ReLU]	0.1341	L_1
6	4.10	[311, 450, 281, 306, 429]	[Sigmoid, Tanh, Sigmoid, ReLU, Tanh]	0.3510	L_1
7	4.70	[163, 307, 437, 279, 427]	[Sigmoid, Sigmoid, ReLU, Sigmoid, Sigmoid]	0.8894	L_1
8	3.78	[358, 175, 145, 494, 138, 264]	[Tanh, Tanh, Tanh, Sigmoid, Sigmoid, Sigmoid]	0.1645	L_2
9	3.90	[170, 320, 433, 271, 116]	[Sigmoid, Sigmoid, Sigmoid, Sigmoid, ReLU]	0.9766	L_2

Table 1: Summary of Hyperparameter Tuning Trials

These hyperparameters formed the final model, which was subsequently trained on the training dataset and validated using the validation dataset to approximate out-of-sample performance. The hyperparameters selected for the final model included:

- Hidden layer sizes: [486, 479, 404]
- Activation functions: [Sigmoid, ReLU, Sigmoid]

- Optimiser: Adam
- Regularisation and regularisation value: L_2 and $\lambda = 0.0721$

The final model was trained over iterations during which the optimiser, Adam, adjusted weights to minimise the loss function, and a learning rate scheduler reduced the learning rate when performance plateaued, further enhancing convergence. During the training process, the model was evaluated on the average loss on the validation data, providing insight into its ability to generalise to unseen data.

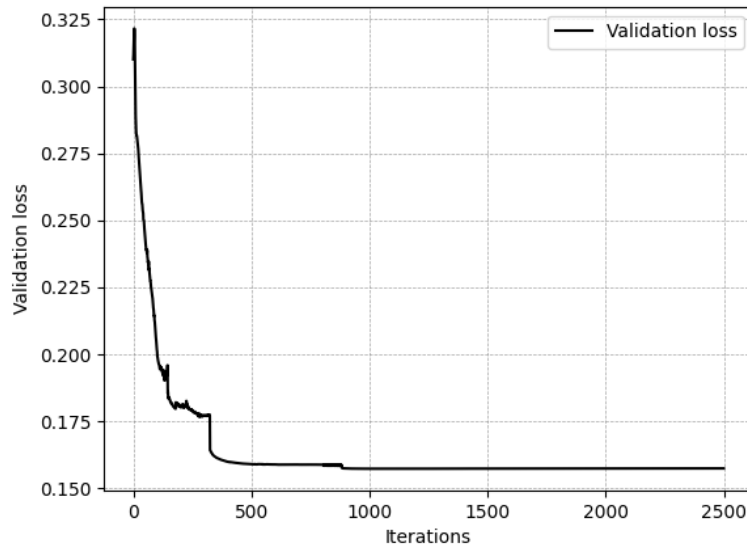


Figure 8: Validation loss

Figure 8 illustrates the learning process of the model where there was a lot of instability within the first 500 iterations, thereafter convergence appears to have been achieved by the gradient descent process.

6.2 Profit and Loss Distribution Comparison

To compare the effectiveness of the Black-Scholes hedge and Neural Network hedge, 10,000 scenarios were simulated. The PNL distribution of the neural network hedge appears to be more concentrated around zero compared to Black-Scholes hedge, with a marginally smaller spread indicating a more stable performance. This suggests that the neural network approach might provide a more effective hedge across a range of scenarios.

To quantify the differences between the two strategies, we compute and compare their mean and standard deviation values. Table 2, shows that the neural network hedge achieves a lower

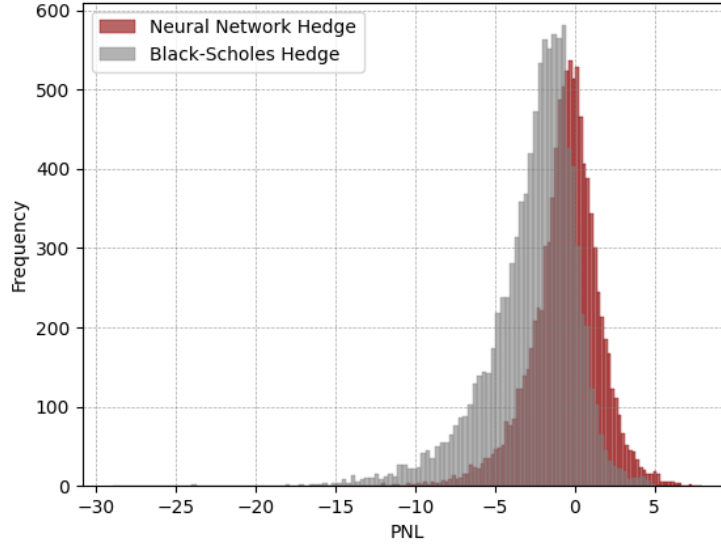


Figure 9: Profit and Loss distribution for Black-Scholes and Neural Network Hedge on 10000 scenarios.

standard deviation than the Black-Scholes hedge, reflecting its ability to provide more consistent outcomes. Additionally, from this table, it is clear the neural network hedge outperforms the Black-Scholes in terms of both the mean and standard deviation of PNL.

Statistic	Black-Scholes Hedge	Neural Network Hedge
Mean PNL	2.646	-0.560
Standard Deviation	2.873	2.187

Table 2: Mean and standard deviation of PNL distributions for Black-Scholes and Neural Network Hedge across 10,000 scenarios.

These findings suggest that on average the neural network hedge performs better compared to traditional Black-Scholes model and with a slightly smaller standard deviation, the party entering into the option assumes a lower risk by utilising neural network hedge.

6.3 Comparison on Observed Stock Prices for June

The final step in our research was to evaluate the hedging performance of both strategies using the actual stock prices observed during the month of June, as illustrated in Figure 10. The observed payoff on the stock prices at maturity, $\Pi = (S_T^1 - S_T^2)^+ = 9.83$, indicates that the holder of Google stock entering into the contract would have a positive payoff at maturity.

6.3 Comparison on Observed Stock Prices for June

Through this period, Amazon's stock price consistently remained higher than Google's, this price divergence plays an important role in hedging performance since the payoff depends on the difference between the two stock prices. In our case, a larger gap in stock prices typically leads to a higher potential payoff, but it increases the uncertainty, which requires more precise hedging adjustments in the attempt to mitigate risk.

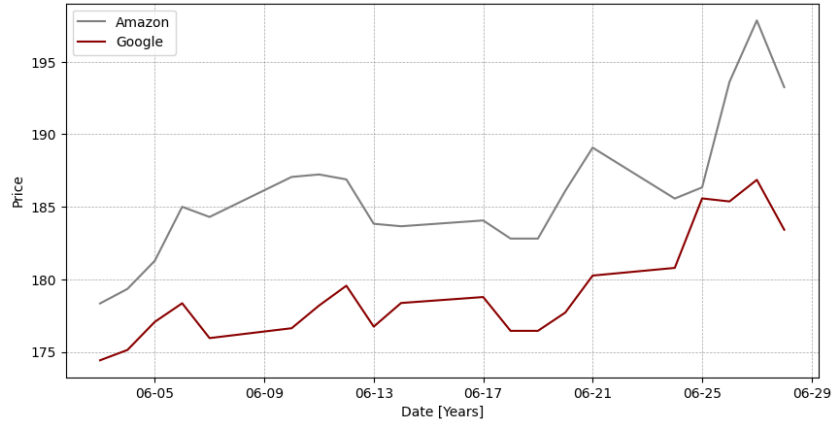


Figure 10: Actual observed June stock prices

Figure 11 shows the portfolio over the course of June, comparing the Black-Scholes hedge and neural network hedge strategies. This demonstrates how each strategy managed the hedging process in response to the movement of the underlying stock prices throughout the month. With Amazon's stock consistently outperforming Google, the neural network hedge proved to be more effective in capturing these market conditions, show more stable performance and fewer variations in portfolio value. This suggests that the neural network approach was better at adapting to increasing payoff potential, driven by Amazon's stronger performance.

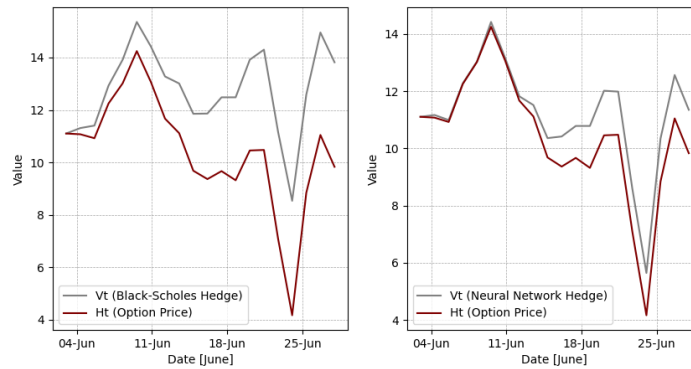


Figure 11: Value of the portfolio for the month of June obtained from both strategies

In contrast, the Black-Scholes hedge, being more rigid and based on the assumption of constant volatility, showed greater sensitivity to price fluctuations. The increased price gap between these two stocks resulted in larger deviations between the option price and portfolio value.

This real-world comparison highlights the practical advantages of the neural network hedge. While both strategies are effective in theory, the neural network approach appears better suited to adapting to complexities of actual market data, including significant price differences between assets.

7 Conclusion

7.1 Summary of Findings

In this project, we have demonstrated the comparative performance of the Black-Scholes model and a neural-network-based approach for hedging a basket swap option involving Amazon and Google stocks. Making use of the Heston model as sample path generator, which effectively captures stochastic volatility, multiple stock price scenarios were simulated over the period of interest for both stocks. The hedge ratios, $(\varphi_t^1, \varphi_t^2)$, to hedge this option were computed using the Black-Scholes framework and a trained neural network model, thereafter profit and loss (PNL) distributions were evaluated based on the different scenarios allowing us to compare the performance of these two hedging strategies.

The neural network hedge exhibited a more stable PNL distribution, with a mean closer to zero and a lower standard deviation than Black-Scholes model. By incorporating stochastic volatility via Heston model and learning a non-linear approach to hedge, the neural network model provided a more stable hedging performance which provided a more effective way to manage risk. The results of this study suggest that neural networks can be a viable alternative for hedging strategies in financial markets, offering flexibility and accuracy in handling market complexities compared to traditional hedging approaches.

7.2 Limitations and Future Research Directions

While this study has shown promising results for neural network-based hedging strategies, several limitations must be acknowledged. First, the neural network model's performance depends heavily on the chosen hyperparameters and network architecture. The process of hyperparameter tuning, while necessary, is time-consuming and computationally expensive, and selecting the optimal configuration can be subjective.

Future research could explore ways to mitigate these limitations. One direction would be to investigate more advanced neural network architectures, such as recurrent neural networks (RNN) or transformers, to capture temporal dependencies in the data. Furthermore, comparing

this neural network-based hedge against other volatility models or using alternative asset classes, such as exotic derivatives, would provide deeper insights into its robustness and applicability.

Appendix

Girsanov's Theorem

Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered space and $W = (W^1, \dots, W^d)$ be a d -dimensional Brownian motion on this space. Let $\gamma = (\gamma^1, \dots, \gamma^d)$ be a vector of progressive processes satisfying

$$\mathbb{P}\left(\int_0^T (\gamma_s^i)^2 ds < \infty\right) = 1, \quad i = 1, \dots, d.$$

Define the continuous local martingale Z by

$$Z_t := \exp\left(\sum_{i=1}^d \gamma_s^i dW_s^i - \frac{1}{2} \int_0^t \|\gamma_s\|^2 ds\right).$$

Assume that Z is a martingale and define a new probability measure $\mathbb{P}^*$ by

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} := Z_T.$$

Then the process $\tilde{W}$ defined by

$$\tilde{W}_t := W_t - \int_0^t \gamma_s ds$$

is a Brownian motion with respect to $\mathbb{P}^*$, thus we can rewrite the SDE for X in terms of $\tilde{W}$ as

$$\begin{aligned} dX_t &= (\mu - r)X_t dt + \sigma dW_t \\ &= X_t((\mu - r)dt + \sigma d\tilde{W}_t + \sigma \gamma_t dt) \\ &= X_t((\mu - r + \sigma \gamma_t)dt + \sigma d\tilde{W}_t) \end{aligned}$$

Hence, for X to be a $\mathbb{P}^*$ local martingale, we need to choose

$$\gamma_t = -\frac{\mu - r}{\sigma}$$

The quantity $(\frac{\mu - r}{\sigma})$ is often called the *market price of risk* or the *Sharpe Ratio*. Choose γ_t as such, the SDE for X becomes

$$dX_t = X_t \sigma d\tilde{W}_t,$$

a local martingale under $\mathbb{P}^*$, in fact, X is a true martingale, so $\mathbb{P}^*$ is a martingale measure. By the martingale representation of Brownian motion, we observe that X satisfies PRP with respect to $\mathbb{P}^*$, hence the market is complete and the ELMM measure is unique.

Python Code

Here are the links to the Python code:

- A.1. Data Generation
- A.2. Parameter Estimation and Hedging
- A.3. Observed Stock Prices June

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